Introduction to Economics

Class 12 — Perfect competition; monopoly begins

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Plan for today

- A firm that cannot choose its price: how much to produce
- When to produce nothing; the firm's supply curve
- A market's supply: a day of electricity prices
- The long run: entry and exit until economic profit is zero
- Monopoly: a first look

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's second class. On Thursday the print shop's costs were taken apart: fixed and variable, the cost per bag, the cost of one more bag. Today those costs meet a price the shop cannot choose, and they decide how many bags it prints, when it prints none at all, and what happens over the years once anyone can open a shop next door. A real day of electricity prices opens the class and is explained after the break; the last two slides before the summary meet the opposite kind of firm, one that sets its own price.

Slide 2 Plan for today

The class at a glance. Before the break, one firm: why the print shop takes its price as given, how it picks the number of bags from the cost of one more, when it is better to print nothing for a while, and how those answers add up to its supply curve. After the break, many firms: a market's supply built from its sellers, with a real day of electricity prices as the test; then what entry and exit do over the years; and, to end, the one seller with no rivals.

Two halves with a ten-minute break.

Warm-up: land, labour or capital?

Our print shop's inputs: land, labour or capital?

1. The workshop building it rents
2. The plot of ground under the building
3. The printers' working hours
4. Who owns and supplies all of these?

In pairs, two minutes: four answers; then hands per item.

Factors of production: land, labour, capital (class 3, slide 22).

Slide 3 Warm-up: land, labour or capital?

A warm-up on the factors of production, with the print shop as the example. Before class, decide for each item whether nature made it, people made it, or it is someone's time, and ask who, in the end, owns or supplies all of them; then ask which everyday word on the list does not mean what economists mean by it.

Revealed in class.

Warm-up: who should make the summaries?

Exam prep: flashcards and summaries

In one hour — Ana: 6 flashcards or 2 summaries · Ben: 3 flashcards or 1 summary

1. Can both gain by specialising?
2. Ben gets faster: 3 flashcards or 2 summaries. Who makes summaries?
3. After 2: at what price could they swap?

In pairs, one minute: three answers; then hands.

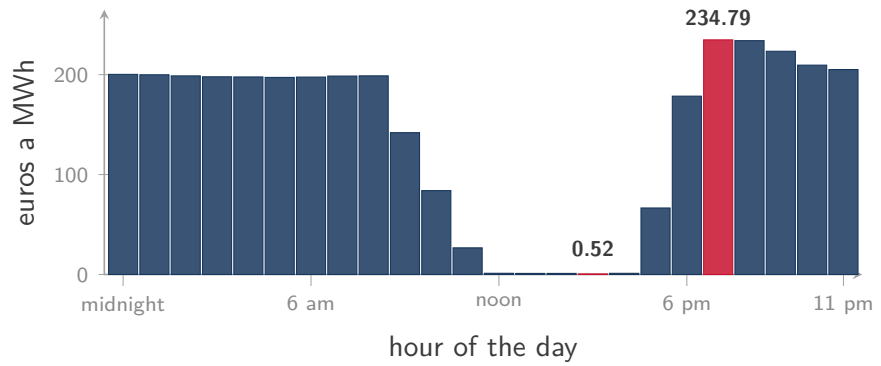
Comparative advantage: the lower opportunity cost (class 3, slide 18).

Slide 4 Warm-up: who should make the summaries?

A second warm-up, on comparative advantage. Before class, work out for each of the two students what one summary costs in flashcards, first with the original numbers and then after the second student speeds up, and ask when the two opportunity costs differ and what range of swap prices would leave both better off.

Revealed in class.

Sunday 11 October 2026: electricity at 0.52 euros a MWh at 3 pm, 234.79 at 7 pm



Same plants, same market, four hours apart: why?

OMIE (the Spanish–Portuguese wholesale electricity market), day-ahead price, mainland Spain, 11 Oct 2026; each bar averages the hour's four quarter-hour prices. 1 MWh = 1,000 kWh.

Slide 5 Sunday 11 October 2026: electricity at 0.52 euros a MWh at 3 pm, 234.79 at 7 pm

The hook: one wholesale electricity market, one Sunday, and a price that was about half a euro per megawatt-hour in mid-afternoon and some 450 times that four hours later, with the same power plants and the same buyers. Your bill at home swings far less: most contracts fix the price, and every bill adds grid charges and taxes on top of this wholesale price. Nobody fixes these numbers: they come out of supply meeting demand in each hour, and the day's tools explain both figures after the break. Keep the question open until then.

Perfect competition: our print shop is a price taker at 4 euros a bag

Tote printing in Barcelona:

- many shops, each one small
- the same printed bag at each
- anyone can open a shop

price taker: no shop can move the price



Each extra bag adds 4.00 to revenue: **marginal revenue** (MR) = price.

Competitive market: many sellers of the same good; none can move the price (class 5, slide 8).

Slide 6 Perfect competition: our print shop is a price taker at 4 euros a bag

Why the shop cannot set its price: with many small shops printing the same bag and nothing stopping a new one from opening, a customer quoted more than the going price walks next door, and a shop asking less only gives money away, since it can already sell all it prints at 4.00. The flat line is the demand this one shop faces, not the market's. The consequence carries the whole class: because the price does not move with the shop's output, each extra bag adds exactly 4.00 to revenue, so marginal revenue equals the price, and the shop's whole decision becomes one comparison, the price against the cost of the next bag. A seller that can choose its price comes at the end of the class.

At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

printers	bags a day	total cost	MC a bag ΔTC ÷ Δbags	revenue 4.00 × bags	profit revenue − cost
0	0	180	—	0	−180
1	40	300	3.00	160	−140
2	120	420	1.50	480	60
3	180	540	2.00	720	180
4	**220**	**660**	**3.00**	**880**	**220**
5	240	780	6.00	960	180

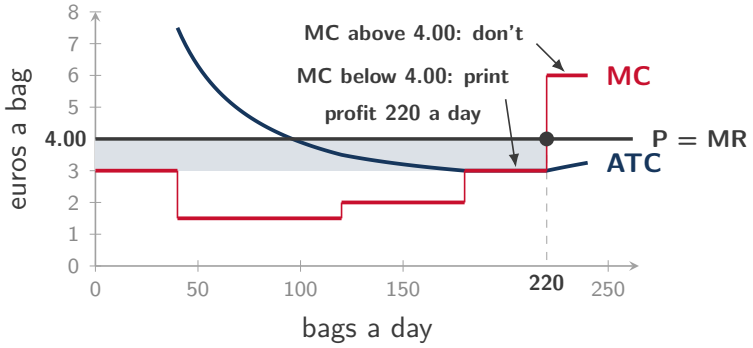
The fifth printer's 20 bags cost 6.00 each and sell for 4.00: profit falls.

Total cost (TC) and marginal cost (MC), the cost of one more bag: class 11, slide 10.
Costs here: those paid out.

Slide 7 At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

Thursday's cost table with revenue and profit added. Profit here is revenue minus the costs actually paid out, so it starts at minus the rent and lease when nobody prints. The MC column, the cost of one more bag, is the reason the peak sits where it does: every bag up to 220 costs at most 3.00 to make and sells for 4.00, so each adds to profit, while the fifth printer's bags cost 6.00 each and lose 2.00 apiece. Not the first row where the cost per bag hits its low of 3.00, not the row with the most bags: the best row is where the next bag stops paying. The owner's own time stays out of this table because it is the same whatever she prints; it returns in the long run.

Print while marginal cost is below the price: 220 bags, profit 220 a day



Profit = (price − ATC) × bags = (4.00 − 3.00) × 220 = 220 a day.

MC: what one more bag adds to total cost · ATC, average total cost: total cost ÷ bags (class 11, slide 10).

Slide 8 Print while marginal cost is below the price: 220 bags, profit 220 a day

The same decision as a picture, and the rule the reading states: produce where marginal cost climbs past the price, which for a price taker means where P = MC, since MR = P. The reason is in the two regions: left of 220 each extra bag adds 4.00 of revenue and less than that of cost, so profit rises; right of it the next bag costs 6.00 and profit falls. What the rule is good for: the shop needs only the price and its own cost of one more bag, nothing about the market as a whole. The box is profit as the reading draws it: height price minus ATC, Thursday's total cost per bag, width the number of bags. Stopping at 180, where ATC first reaches its low of 3.00, would leave 40 bags unprinted that each earn 1.00.

In pairs, three minutes: a strawberry farm at the market price

A Maresme farm sells strawberries at the market price.
Extra cost of each further 100 boxes a day:

boxes a day	0→100	100→200	200→300	300→400	400→500
euros a box	4	5	7	10	14

1. Price 8 euros a box: how many boxes a day?
2. The price rises to 12: how many now?
3. At 8 euros, what do boxes 301–400 do to profit?

One answer per question; then hands.

Slide 9 In pairs, three minutes: a strawberry farm at the market price

Exercise 1 is slide 8 on a farm, with the cost of each further hundred boxes in a table. Before class, for each price run down the table and ask which hundreds are worth picking and where to stop; then ask what a hundred boxes that cost more than they sell for do to the farm's profit.

Revealed in class.

Below 2 euros a bag the shop prints nothing: the rent is paid either way

price a bag	**4.00**	**2.50**	**1.80**
best if open	220 bags	180 bags	120 bags
revenue	880	450	216
printers' wages	480	360	240
rent and press	180	180	180
profit if open	+220	−90	−204
profit if shut down	−180	−180	−180
so	print	print	**shut down**

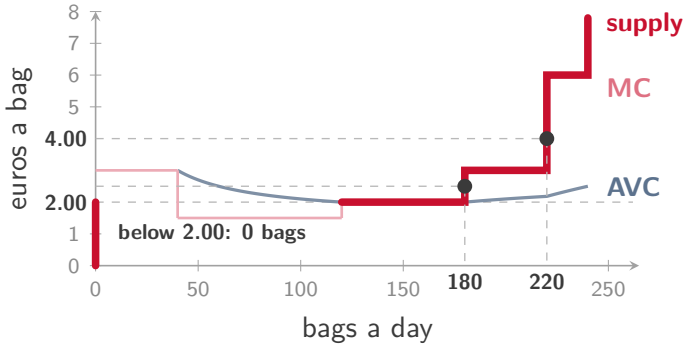
Shut down (for now) if the price is below AVC: here below 2.00.

AVC, average variable cost: printers' wages ÷ bags, lowest 2.00 · fixed: 180 whatever the output (class 11, slides 9–10) · sunk: already paid, gone whatever you choose (class 2, slide 23).

Slide 10 Below 2 euros a bag the shop prints nothing: the rent is paid either way

Each column first finds the best output by slide 8's rule and then compares two losses, open and shut down. The rent and lease, 180 a day, are owed either way this month: they are sunk, paid whatever she decides, so they cannot settle the question; what can is whether revenue covers the printers' wages, the variable cost. At 2.50 it does, with 90 to spare toward the rent, so staying open loses less than shutting down; at 1.80 it does not. Divided by bags, that is the rule: shut down when the price is below average variable cost, the wages per bag, whose lowest value is 2.00. It is why a hotel stays open at a loss in a quiet month.

The shop's supply curve: its rising marginal cost above the lowest AVC



A higher price, more bags: supply slopes up because MC rises.

AVC: printers' wages ÷ bags; MC: the cost of one more bag (class 11, slides 10 and 16).

Slide 11 The shop's supply curve: its rising marginal cost above the lowest AVC

Slides 8 and 10 drawn as one curve. At each price the shop prints where its MC staircase, the cost of one more bag, reaches the price, but only if the price is at least the lowest AVC, the wages per bag; below 2.00 it prints nothing, so the supply curve runs up the price axis and then along the staircase. The faint part of MC, where it is not above AVC, is never used. This is why class 5's supply curve slopes up, and what it is made of: the sellers' marginal costs, so anything that raises the cost of one more unit, wages, energy or a tax per unit, shifts supply up. The two dashed prices are slide 10's answers read off the curve.

A food truck's festival week: open today? (optional)

Pitch for the week: 600, paid, no refund · each day it opens: ingredients and a helper, 250

1. Sunday's sales would be 300. Open?
2. Rain: Monday's sales would be 200. Open?
3. Does the 600 change either answer?

In pairs, two minutes: three answers.

Slide 12 A food truck's festival week: open today? (optional)

An optional exercise on slide 10's rule with an everyday firm. Before class, list which of the truck's costs change with the decision to open on a given day and which do not, and compare each day's sales with the first kind only.

Revealed in class.

Break

Ten minutes

Problem set 2: due Thursday 5 November, on paper

slide 13 of 25

Slide 13 Break

Ten minutes. Problem set 2 is due on Thursday 5 November, on paper, in class. After the break: the electricity question, then many firms instead of one.

Each plant runs while the price covers its extra cost of a MWh

	Solar farm	Gas plant
extra cost of a MWh	about 0	70
3 pm: price 0.52		
7 pm: price 234.79 a little sun left		

Panels, turbines, loans: paid either way, so they do not decide.

In pairs, one minute: four answers, one per cell.

70 a MWh: gas plus the CO_2 permit (class 11, slide 14) · slide 10's rule: price against average variable cost; here every MWh costs the same, so that is the extra cost of a MWh.

slide 14 of 25

Slide 14 Each plant runs while the price covers its extra cost of a MWh

The first half of the electricity question. A power plant sells into a competitive market, one price for every MWh that hour, so a rule from before the break applies to it as it stands. Before class, decide for each cell which cost each plant should compare with the hour's price, and what the position of the sun does to what each can sell.

Revealed in class.

Back to Sunday 11 October: 3 pm and 7 pm

Revealed in class

Try it: at 3 pm the sun is high; by 7 pm it is setting. Which plants does Spain need at each hour?

Slide 15 Back to Sunday 11 October: 3 pm and 7 pm

The answer to slide 5, built in class from slide 14's plants and the market supply curve of class 5. Before class, ask what changes between mid-afternoon and early evening in what can be offered and in what is wanted, and how a market's supply at one price is put together from its sellers.

Revealed in class.

In pairs, three minutes: three egg farms, one market

Each farm's cost per dozen stays the same up to its limit.

	cost per dozen	dozen a day (at most)
Farm A	1.00	300
Farm B	1.60	200
Farm C	2.20	100

1. Market supply at 2.00 a dozen?
2. At 2.50?
3. At 1.20?
4. Buyers want 550 dozen a day at any price. The price?

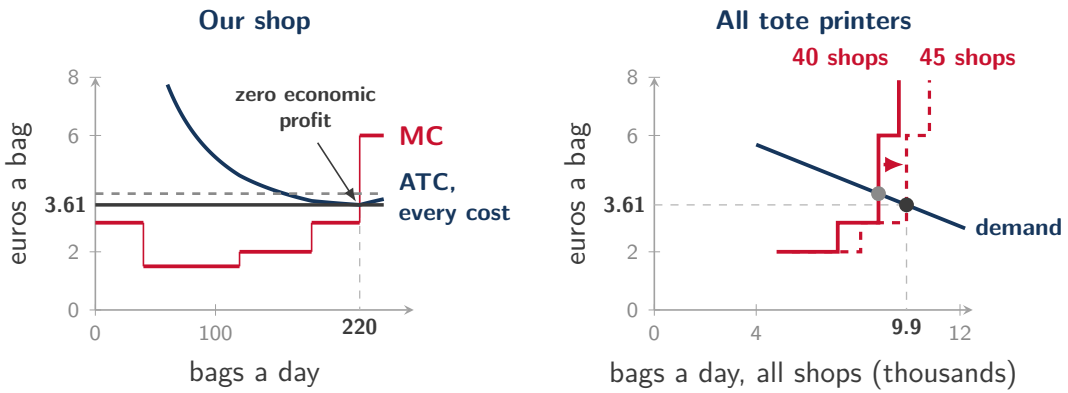
One answer per question; then hands.

Slide 16 In pairs, three minutes: three egg farms, one market

Exercise 2 is slide 15 with three farms. Before class, decide at each price which farms produce at all, then ask what has to be added to get the market's quantity and what must not be; for the last item, ask who has to be producing for the buyers to get what they want, and what that means for the price.

Revealed in class.

Profit draws new shops: the price falls until economic profit is zero



Entry stops when price = lowest ATC (every cost counted): 3.61.

Each shop: one press · every cost: class 11's costs plus the owner's 135 a day (class 11, slide 6)

long run: time for every cost to change (class 11, slide 18).

Slide 17 Profit draws new shops: the price falls until economic profit is zero

The long run begins, with every cost counted: Thursday's costs plus the owner's own 135 a day, the design job and the interest her savings gave up, which is what separates accounting from economic profit. Counting them lifts ATC; its lowest point, 3.61, is the price at which the shop pays her exactly what her best alternative would. At 4.00 she earns more, and because anyone can open a shop that gap is an invitation: new shops add their supply, the market curve moves right and the price falls until the gap is gone. Firms enter while the price is above ATC and exit while it is below, so the process can only stop where price equals ATC; since each shop also prints until its MC climbs past the price, that is the lowest ATC. The use of the result: in a competitive industry a profit is copied away, so a firm that wants to keep one must keep rivals out.

In the long run the price sits at the lowest ATC: market supply is flat at 3.61



More demand brings more shops, not a higher price.

Zero economic profit: the owner's accounts still show 2,700 a month — what her job and savings would pay.

Slide 18 In the long run the price sits at the lowest ATC: market supply is flat at 3.61

Where the price comes to rest. Any price above 3.61 is a profit someone will open a shop to take, and any price below it a loss that makes someone give up the lease and the press, so in the long run the price can only sit at the lowest ATC with every cost counted: the long-run market supply is flat there. More demand lifts the price for a while, as each shop climbs its own supply curve, but then brings new shops rather than a lasting rise; less demand makes some leave. Zero economic profit is not zero income: the owner covers every cost, her own time included, and her accountant still shows the 2,700 a month her job and savings would have paid, so she stays. Where firms' costs differ, as with the power plants, the long-run supply slopes up instead.

In pairs, three minutes: repair stalls in the long run

Phone-screen repair stalls, all alike. Lowest ATC, the owner's time counted: 40 euros a repair. Price today: 55.

1. Economic profit today: positive, zero or negative?
2. Where does the price go?
3. At 40, do the owners quit?
4. A new 5-euro fee on every repair. Long-run price?

One answer per question; then hands.

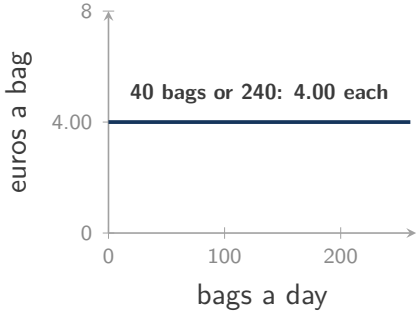
Slide 19 In pairs, three minutes: repair stalls in the long run

Exercise 3 is slides 17 and 18 on a new trade. Before class, compare the price with the lowest cost per repair when the owner's time is counted and ask what that comparison sets in motion, where it ends, and how the owners fare at the end; then ask what a fee on every repair does to that lowest cost and so to the long-run price.

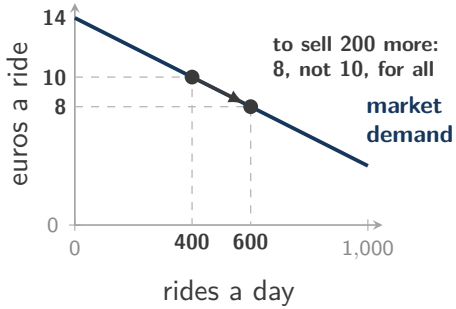
Revealed in class.

A monopoly is the only seller: to sell more, it must cut its price

Our print shop: a price taker



A mountain's only cable car: a price maker



A price taker sells any amount at the price; a monopoly picks a point on the demand curve.

Which point it picks: Monday 9 November (reading Ch. 11).

Slide 20 A monopoly is the only seller: to sell more, it must cut its price

The print shop faces a flat demand because its customers have many other shops to go to; a seller with no rivals and no close substitute faces the whole market's demand, which slopes down. That is why it must cut its price to sell more, and for every rider, not only the extra ones: as the only seller, it is the market. So it chooses a point on the demand curve, a price and the quantity that goes with it, which makes it a price maker; what that is good for is that no rival pushes its price down to cost, as entry did on slide 17. It still cannot charge what it likes and sell the same amount: a higher price means fewer buyers. Which point it picks, and what that costs society, is Monday's question.

Rivals are kept out by a key resource, a law, or costs that fall with size

A key resource

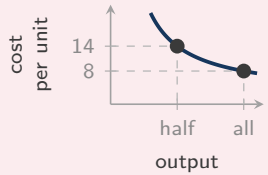
the only quarry of a rare stone

The law

a patent: no copies for 20 years

Falling costs: a natural monopoly

Spain's high-voltage grid: one company, Red Eléctrica



own numbers: one firm, 8 a unit; two firms, half each, 14

What keeps rivals out is a **barrier to entry**.

Economies of scale: ATC falls as output grows (class 11, slide 20).

European Patent Convention, art. 63 (20 years from filing); Ley 24/2013 del Sector Eléctrico, art. 34.2 (Red Eléctrica, transportista único).

Slide 21 Rivals are kept out by a key resource, a law, or costs that fall with size

Why a monopoly can last: something must stop rivals from entering, otherwise slide 17's entry would push the price down to cost here too. The three cards are three of the reading's four sources (the fourth: buying up rivals): one firm owns the input everyone needs; the law grants one firm the sole right, as a patent does for twenty years so that inventing pays; or one firm is simply the cheapest way to serve the whole market, because its cost per unit keeps falling as output grows, Thursday's economies of scale, so a second network would double the fixed cost for the same electricity. The grid that slide 15's competing plants share is such a natural monopoly, which is why grids, pipes and rail tracks are regulated rather than split. Size alone is not the test: a monopoly is a seller without a close rival.

Matcha gets popular: which curve moves? (optional)

Barcelona's matcha cafés: many, all alike; lowest ATC 3.50 a cup.

1. A video makes matcha fashionable. Which curve moves?
2. The price rises to 4.50. Does each café's supply curve shift?
3. New cafés open. What moves now?

In pairs, two minutes: three answers.

Slide 22 Matcha gets popular: which curve moves? (optional)

An optional exercise on the difference between a shift and a move along a curve, using slides 17 and 18. Before class, for each event ask whether the quantity people want or firms offer changed at every price, or whether only the price changed; then ask what the arrival of new cafés does to the market and to the price.

Revealed in class.

One grid or two: positive or normative? (optional)

1. *"Two grids side by side would cost more per MWh carried."*
2. *"The grid should belong to the state."*
3. A and B both want cheap electricity. A: *"a state company would keep costs lowest."* B: *"a regulated private one would."* Values or a prediction?

In pairs, two minutes: three answers.

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

Slide 23 One grid or two: positive or normative? (optional)

An optional exercise on positive and normative claims, hung on slide 21's grid. Before class, ask of each claim whether evidence could in principle settle it or whether it rests on what someone thinks ought to be; for the third item, ask what exactly the two people disagree about, given that they share a goal.

Revealed in class.

A price taker produces while marginal cost is below the price.

Below the lowest average variable cost, it produces nothing: fixed costs are sunk.

Market supply adds the firms' quantities; in the long run economic profit is zero.
A monopoly is the only seller: to sell more, it must cut its price.

Slide 24 In one line

The class in four lines, to read slowly: the first is slides 6 to 9, the second slides 10 and 11, the third slides 14 to 19, and the fourth slides 20 and 21.

Before Thursday 5 and Monday 9 November

- Thursday 5 November: poster session, the full two hours; one presenter per group, the whole group answers questions
- Bring your poster printed, A2 portrait; e-mail the PDF by Wednesday 4 November, 8 pm
- Problem set 3 (Block 4): handed out today; due Thursday 12 November, on paper, in class
- Monday 9 November: monopoly; oligopoly and the prisoner's dilemma — read Mankiw & Taylor Ch. 11, pp. 242–261 and Ch. 13, pp. 280–296
- Practise in Sextant: the units the course page lists



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Slide 25 Before Thursday 5 and Monday 9 November

Thursday 5 November is the poster session, the full two hours: one presenter per group, the whole group answers questions; bring the poster printed, A2 portrait, and e-mail the PDF by Wednesday 4 November, 8 pm. Problem set 3, on Block 4, is handed out today and due on Thursday 12 November, on paper, in class; Problem set 2 is due on Thursday 5 November, on paper, at the poster session. Monday 9 November: monopoly, then oligopoly and the prisoner's dilemma. Reading: Mankiw & Taylor chapter 11, pages 242 to 261, and chapter 13, pages 280 to 296. Practise in Sextant with the units the course page lists.

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